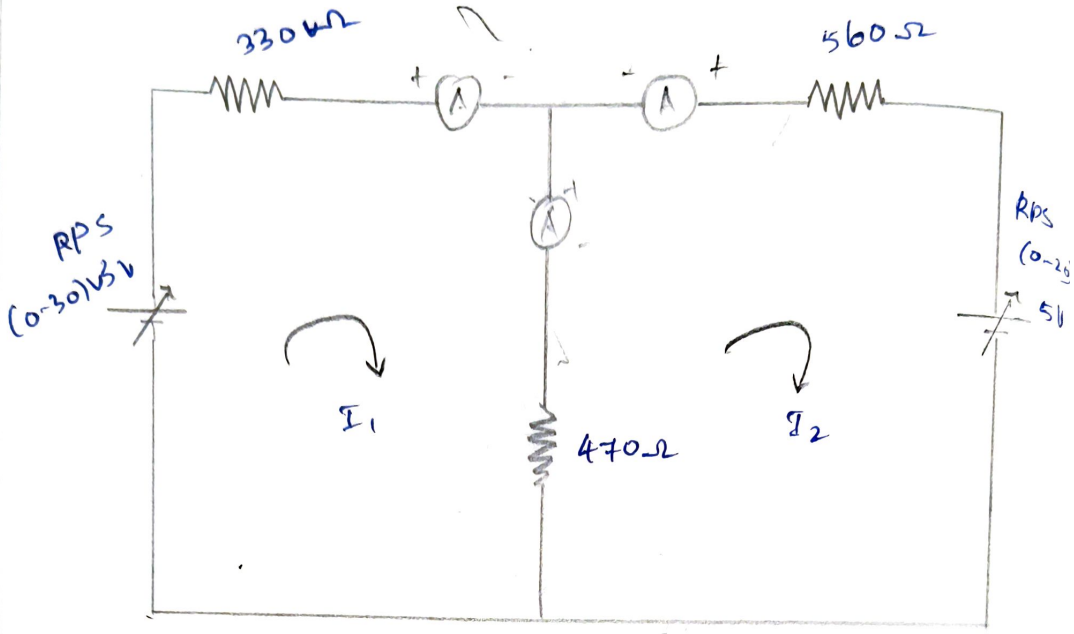
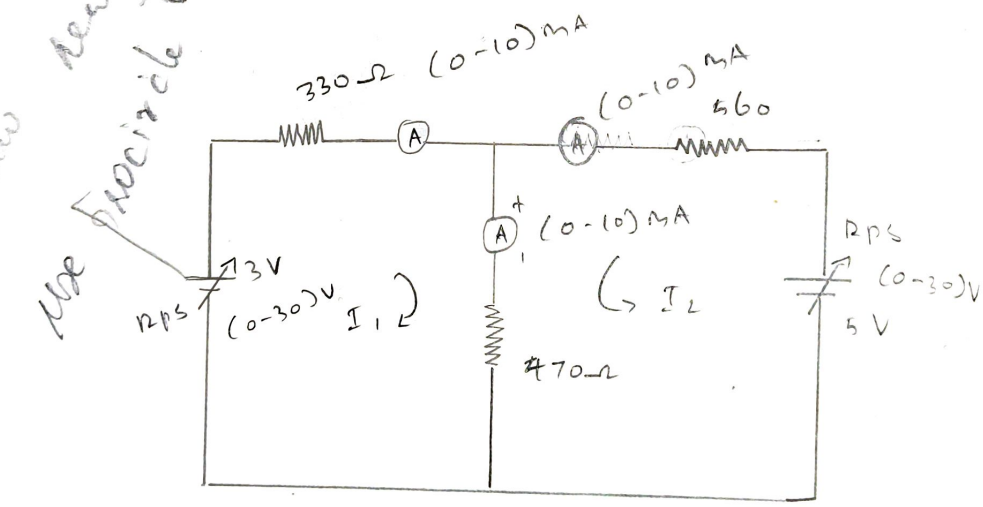
Circuit Diagram:



Draw neatly
use Protractor



Exp. No: 2

Date : 1/02/2025

Determination of current In circuit using Mesh Analysis

Aim:

* To determine the current in circuit using mesh analysis both theoretically and pratically for a given DC circuit.

Apparatus Required:

SL.No	Apparatus	Specification	Quantity
1.	Regulated power supply	(0-30V)	1
2.	Ammeter	(0-10mA) MC	3
3.	Resistors	560Ω, 330Ω, 470Ω,	Each 1
4.	Bread board	-	1

Theory:

* multi-source DC circuit may be analyzed using a mesh current technique. the process involves identifying minimum number of small Loops such that every component exist in at Least one Loop. KVL is then applied to each Loop. the Loop current are referred to as mesh current as each current interlocks or meshes with the surrounding Loop Currents. as as a result ther will be a set of simultaneous equations created an unknown

Tabular column:

parameters	theoretical	practical
I_1	1.22 mA	1.24 mA
I_2	4.29 mA	4.17 mA
$I_1 - I_2$	5.51 mA	5.44 mA

Calculations:

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$R_{11} = 330\Omega + 470\Omega$

$R_{11} = 800\Omega$

$R_{21} = 470\Omega$

$R_{18} = -470\Omega$

$R_{22} = 560\Omega + 470\Omega$

$R_{22} = 1030\Omega$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$\Delta = 800 \times 1030 - (-470)^2$

$\Delta = 824000 - 220900$

$\Delta = 603100$

$$\Delta_1 = \begin{bmatrix} 3 & -470 \\ 470 & 1030 \end{bmatrix}$$

$\Delta_1 = 3090 - 2350$

$\Delta_1 = 740$

$$\Delta_2 = \begin{bmatrix} 800 & 3 \\ -470 & -5 \end{bmatrix}$$

$\Delta_2 = -4000 + 1410$

$\Delta_2 = -2590$

$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{603100} = 1.22\ mA$, $I_1 - I_2 = 1.22 + 4.29$

$I_2 = \frac{\Delta_2}{\Delta} = \frac{-2590}{603100} = -4.29\ mA$, $I - I_2 = 5.51\ mA$

mesh current for each loop once the mesh currents are determined, various branch currents and component voltages may be derived.

Procedure:

* Give connections as per the circuit diagram
* Switch on the supply, vary the RPS and set a particular input voltage
* Note down the readings of ammeters and voltmeters and tabulate them.
* vary the RPS for different input voltages and note down the readings of all the meters
* Reduce the RPS to its minimum value and switch OFF the supply
* using the tabulated, verify Kirchoff's law practically, and verify it theoretically

S.No	Description	Mark Allotted	marks obtained.
1.	Experimental set up connection	20	
2.	observation/readings	20	
3.	calculations/results	20	
4.	viva	20	
5.	record.	20	

Result:

* Thus, the mesh analysis is verified practically and theoretically. the resultant currents for 3V, 5V supply are:

a, the current across 330Ω resistor is 1.2 A

b, the current across 560Ω resistor is 4.2 A

c, the current across 470Ω resistor is 5.3 A